

Breast Cancer Classification using ANN

B.Tech AI Mini Project

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Workflow

- Classify breast tumors as Malignant (0) or Benign (1)
- Use a simple Artificial Neural Network (ANN)
- Dataset: Breast Cancer Wisconsin (569 samples, 30 features)
- Tools: Python, TensorFlow/Keras, scikit-learn

Dataset Information

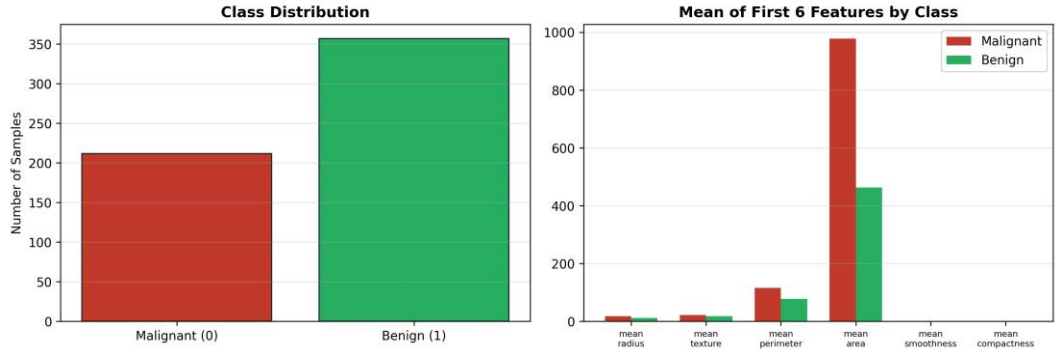
Breast Cancer Wisconsin Dataset Information

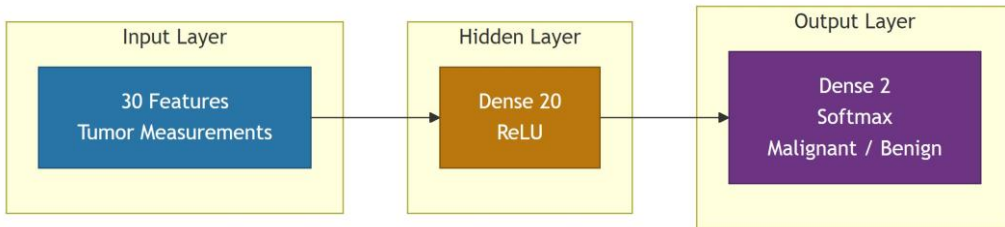
```
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Total samples      : 569
Number of features  : 30
Feature type       : Numeric tumor measurements
Classes           : 0 = Malignant, 1 = Benign
Class counts (full) : Malignant=212, Benign=357

Training samples   : 455
Test samples       : 114
Train labels shape : (455,)
Test labels shape  : (114,)

Preprocessing      : StandardScaler (fit on train only)
Train/Test split   : 80% / 20% (stratified)
TensorFlow version : 2.21.0
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Breast Cancer Dataset Overview





Model Summary

Model: "breast_cancer_ann"

Layer (type)	Output Shape	Param #
hidden_layer (Dense)	(None, 20)	620
output_layer (Dense)	(None, 2)	42

Total params: 662 (2.59 KB)

Trainable params: 662 (2.59 KB)

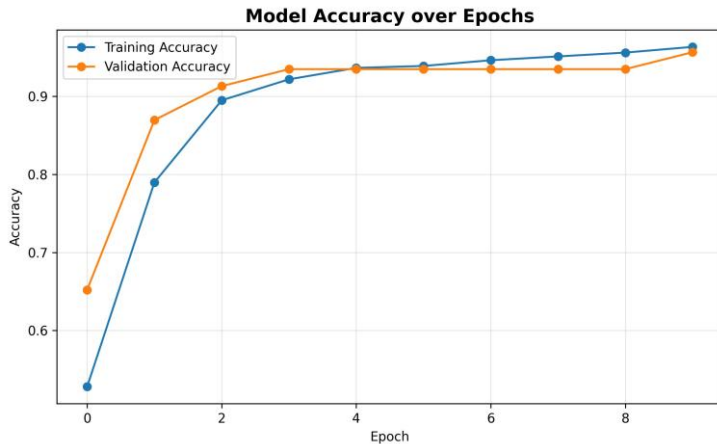
Non-trainable params: 0 (0.00 B)

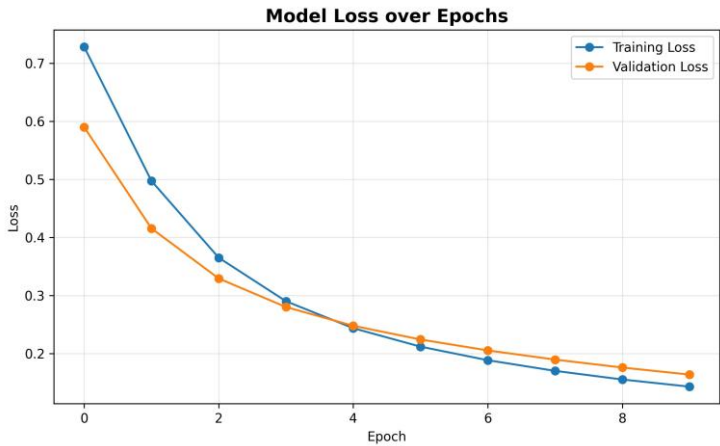
Training Progress (10 Epochs)

Epoch	Train Acc	Val Acc	Train Loss	Val Loss
1	52.81%	65.22%	0.7284	0.5902
2	78.97%	86.96%	0.4975	0.4157
3	89.49%	91.30%	0.3653	0.3295
4	92.18%	93.48%	0.2903	0.2803
5	93.64%	93.48%	0.2438	0.2480
6	93.89%	93.48%	0.2121	0.2244
7	94.62%	93.48%	0.1887	0.2055
8	95.11%	93.48%	0.1703	0.1897
9	95.60%	93.48%	0.1555	0.1760
10	96.33%	95.65%	0.1431	0.1639

Final Training Accuracy : 96.33%

Final Validation Accuracy : 95.65%







Test Accuracy

Test Set Evaluation

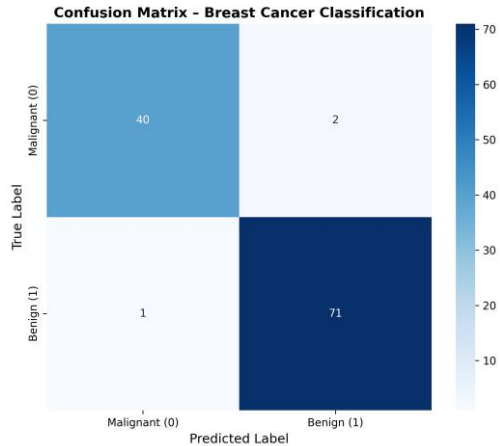
=====

Test Loss : 0.1077

Test Accuracy : 97.37%

Correct predictions : 111 / 114

Misclassifications : 3



Classification Report

	precision	recall	f1-score	support
Malignant (0)	0.9756	0.9524	0.9639	42
Benign (1)	0.9726	0.9861	0.9793	72
accuracy			0.9737	114
macro avg	0.9741	0.9692	0.9716	114
weighted avg	0.9737	0.9737	0.9736	114

Prediction Output with Confidence Scores

Predictive System Output

BREAST CANCER PREDICTION RESULT

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Predicted class index : 1

Predicted diagnosis : Benign

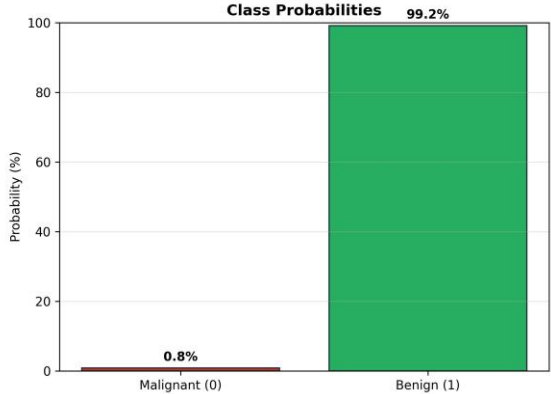
Confidence : 99.15%

P(Malignant = 0) : 0.85%

P(Benign = 1) : 99.15%

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Result: The tumor is predicted to be BENIGN.





- Built a compact ANN for binary breast cancer classification
- Applied stratified split and feature standardization
- Evaluated with accuracy, confusion matrix, and classification report
- Demonstrated a confidence-aware predictive system

Educational demo only — not a medical diagnostic tool.

ThankYou

Questions?